

Exploratory Data Analysis (EDA) report

1.Dataset Description:

The Objective of this analysis is to understand the relationship between Bitcoin trader performance and market sentiment

Two Dataset are combined

2.Data understanding:

The Merged dataset contains trading information together with market sentiment for each trading day.

3. Dataset Shape:

The merged dataset contains X rows and Y columns, indicating sufficient observations for statistical analysis and machine learning.

4. Column Information: Explain in report:

- Total columns
- Numerical features
- Categorical features
- Datetime features
- Memory usage

5. Missing Value Analysis:

1.No missing values were found in the merged dataset. Therefore, no imputation or deletion techniques were required before analysis.

2. Missing values were handled using appropriate preprocessing techniques before further analysis.

6. Statistical Summary:

The statistical summary provides

- Count
- Mean

- Standard deviation
- Minimum
- Maximum
- 25th percentile
- Median
- 75th percentile

The Closed PnL variable exhibits considerable variation, indicating the presence of both profitable and loss-making trades. Trade sizes also vary significantly, suggesting diverse trading strategies among traders.

7.Data Types:

Numerical variables used for statistical analysis and model training , while categorical variables requires encoding during machine learning .

8. Outliers Analysis :

- Method used in Boxplot and IQR Method
- Extreme profit and loss values were identified using the Interquartile Range (IQR) method. These outliers represent unusually large gains or losses and should be carefully considered before model development.

9.Key Findings from Univariate Analysis :

- No missing values were observed in the dataset.
- Numerical variables exhibit high variability.
- Closed PnL contains several extreme outliers.
- Fear and Greed classes are reasonably balanced.
- Long and Short trades are both well represented.
- Bitcoin is the most frequently traded cryptocurrency.